

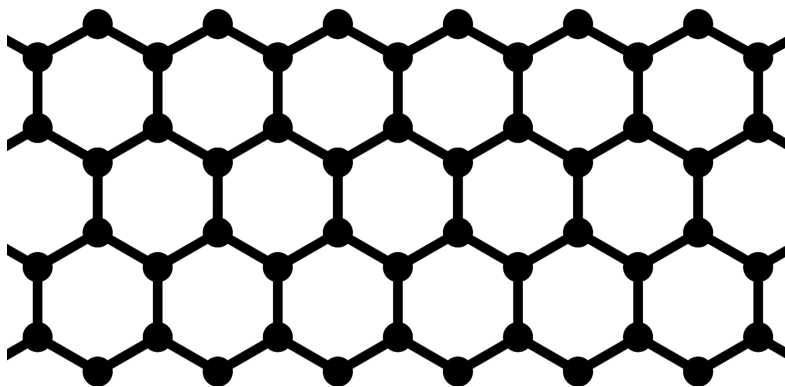
Predictive Procurement: Advanced Time Series Forecasting Methods for Inventory Management

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 Project Repository



Contents

1	Executive Summary	2
2	Introduction	3
3	Methodology	3
4	Results	4
4.1	Book Selection	4
4.2	STL Imputation	5
4.3	Diagnostic Checks	5
4.3.1	ACF and PACF	5
4.3.2	ADF Stationarity	5
4.4	Model Performance	6
4.5	Monthly Sampling Frequency	8
5	Conclusion	8

1 Executive Summary

Retailers and publishers depend on accurate demand forecasts to know how much stock to hold and when to reorder, since errors mean lost sales or wasted inventory. This is compounded by strong seasonal patterns, such as holiday and back-to-school spikes, layered atop long-term trends and anomalous periods such as the COVID-19 pandemic. This forecasting problem is shared across many domains, such as stock and oil price prediction, inventory optimisation, and energy grid demand and flexibility needs.

Drawing on two decades of sales data for a selection of two books with contrasting profiles, this project built and compared a family of forecasting pipelines. A decomposition-based technique first repaired the pandemic gap, separating trend and seasonal rhythm from noise so missing weeks are filled whilst respecting normal seasonal behaviour, rather than naively connecting the dots. Three forecasting approaches are used: a classical statistical method, a gradient-boosted machine learning model, and a deep learning neural network, each automatically tuned. These were compared against each other and then combined in advanced hybrid models to allow the strengths of one method correct another's blind spots.

When looking at single approaches, the classical statistical method proved the strongest and most interpretable, though no one architecture won consistently. Hybrid architectures delivered the best all-round performance by dynamically blending the strengths and weaknesses of the individual methods, tailored to each title's unique pattern.

Deep learning models produced variable results between runs, underscoring the need to train multiple times before trusting any single result. The analysis also confirmed weekly, rather than monthly, tracking is essential for reliable forecasts within the publishing industry. Together, these findings give retailers a practical and repeatable way to separate genuine demand shifts from seasonal noise, supporting smarter procurement, better stock availability, and lower inventory costs.

2 Introduction

Nielsen’s BookScan tracks book sales via ISBNs across retailers, providing comprehensive transactional data to derive valuable business insights. Accurate demand forecasting is vital for stock optimisation and revenue prediction, though it is routinely complicated by long-term seasonality and trend effects. Therefore, models are developed to interpret past sales data to forecast future sales patterns, enabling proactive, data-driven decisions about restocking and procurement.

This work leverages robust time series techniques through reusable pipelines. Transformations are applied for STL decomposition to understand seasonality, trends, and to impute anomalous periods caused by COVID-19 closures, ensuring model generalisability. Three forecasting architectures are compared: classical Auto-ARIMA, machine learning based grid-searched XGBoost, and a deep learning approach with KerasTuner-optimised LSTMs. Furthermore, these architectures are combined into advanced parallel and sequential hybrid models to harness the individual strengths of each.

3 Methodology

Raw sales and ISBN data, originally fragmented across eight genre-based tables, is programmatically ingested, transformed, and concatenated. First, data type coherence was assessed, and then data was resampled weekly, replacing missing weeks with zero sales. ISBNs with recent sales were visualised, with two arbitrarily chosen for thorough analysis based on clear seasonality, consistent 20-year sales volumes, and varied expected trends: *The Very Hungry Caterpillar* and *The Alchemist*.

These were assessed via ADF, ACF, and PACF to inspect stationarity, lag relationships, and verify seasonality. Subsequent models were evaluated across an eight month forecast horizon using a 12-year subset of recent data.

A multi-paradigm forecasting pipeline was implemented:

Modelling Approach	Technique Used
Statistical Baseline	Auto-ARIMA capturing linear trends and 52-week seasonal dynamics using <code>pmdarima</code> .
Machine Learning	XGBoost regression pipeline implemented via <code>sktime</code> , tuning lag windows via expanding-window cross-validation.
Deep Learning	Multi-layer LSTM networks built in <code>TensorFlow/Keras</code> using a 52-week lookback, optimised using <code>KerasTuner</code> .
Hybrid Modelling	Sequential (LSTM forecasting SARIMA residuals) and Parallel (error-optimised weighted averaging) frameworks.

Table 1: Table of modelling approaches employed in this work.

Performance was benchmarked using Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE). MAE quantifies the expected forecast error in terms of sales value, whilst MAPE provides an interpretable, scale-independent percentage. This combination is essential for meaningful comparison of forecast reliability across models and books with different baseline sales.

4 Results

Extended visualisations are available in the associated `.ipynb` notebook.

4.1 Book Selection

Initially, all books with recent sales were visualised to interpret common features. Most books showed strong annual, genre-related seasonality. Educational books saw broader, bi-annual spikes aligning with standard examination periods, others showed strong spikes around the festive period. Many ISBNs saw sales decay to near-zero as consumer popularity subsided or new editions under different ISBNs were released. Future work could apply vector embeddings and similarity scores to title clusters to better understand these decay mechanisms.

The Very Hungry Caterpillar and *The Alchemist* are chosen for further analysis since they exhibit excellent features for testing architectures, showing clear seasonality and consistent sales without inactive periods. This demonstrates maintained popularity and makes them good candidates for forecasting as continued sales are expected, as seen in Figure 1.

In the most recent 12 years, *The Very Hungry Caterpillar* shows a positive sales trend whereas *The Alchemist* maintains a more constant baseline.

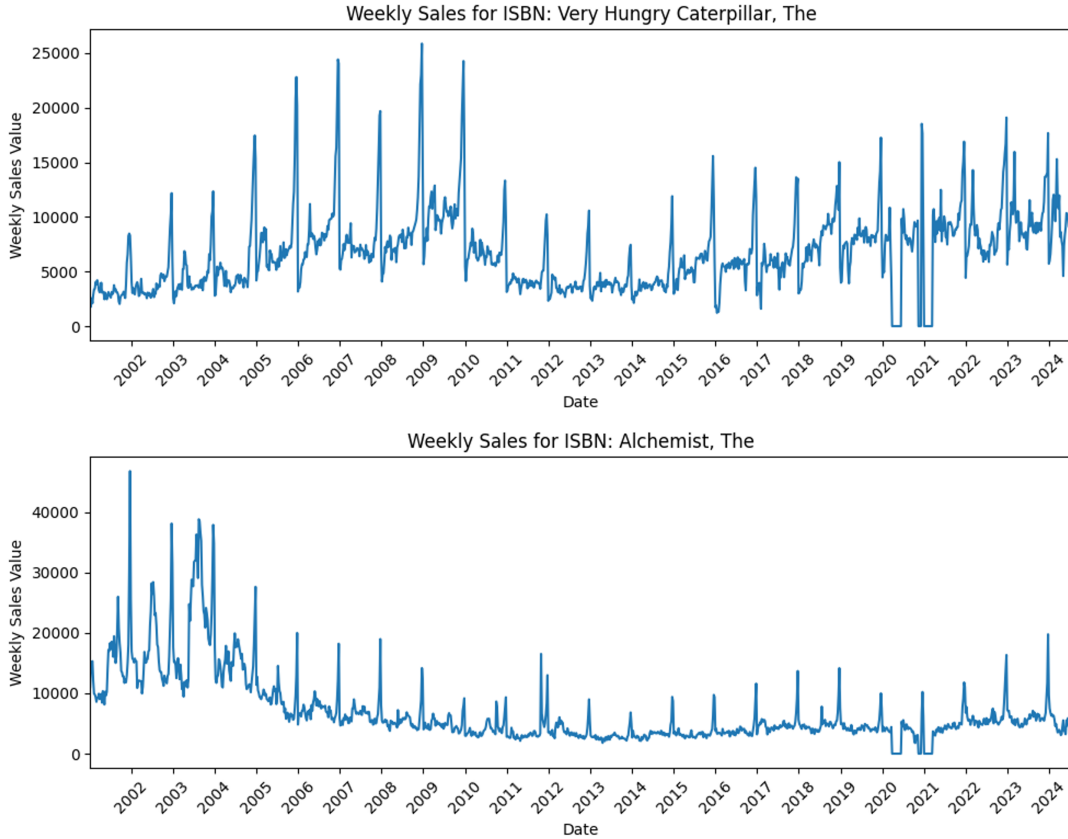


Figure 1: Time series data for *The Very Hungry Caterpillar* and *The Alchemist* across all available data.

4.2 STL Imputation

Around 2020–2022, **all** books show zero sales due to COVID-19 closures. This recent anomaly degraded forecasting performance, so an advanced, season-aware STL imputation method was applied, as visualised in Figure 2.

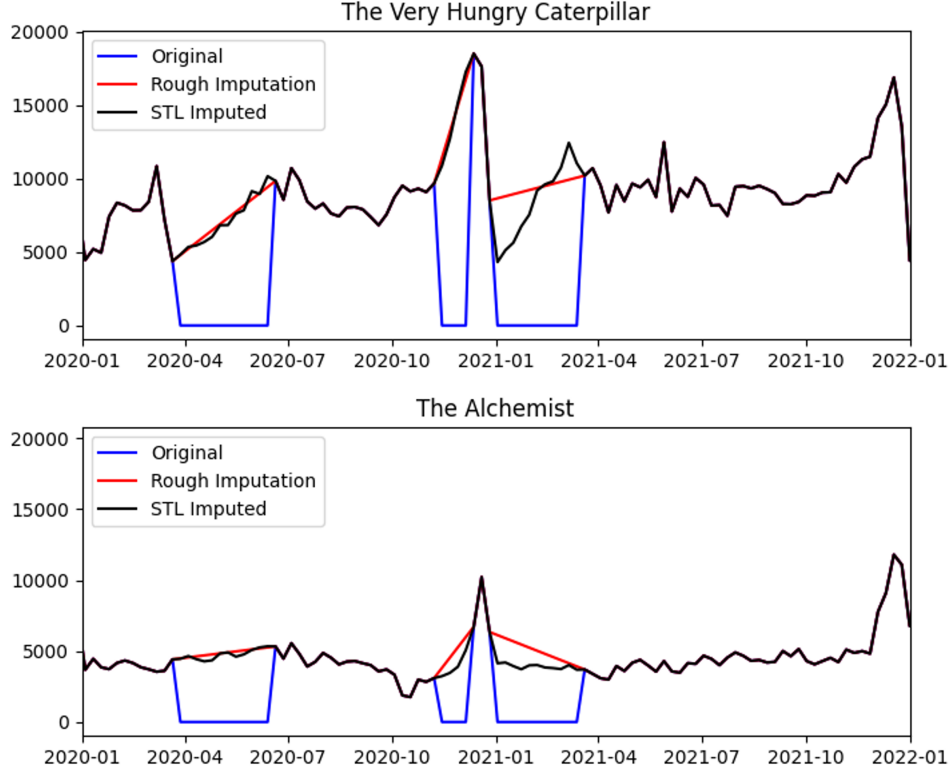


Figure 2: STL-imputed time series used for subsequent forecasting, focussed around COVID-19 closure periods. Blue line shows where sales data is missing, and the red line shows a rough linear imputation where non-zero sales are connected by a straight line. STL is performed on this roughly imputed series to extract trend and seasonal components so a more advanced imputation can be applied to the missing regions, with the final data shown by the black line.

4.3 Diagnostic Checks

4.3.1 ACF and PACF

After imputation, the ACF and PACF plots, Figure 3, confirm that both books exhibit strong yearly seasonality with lag-52 spikes alongside lag-1 correlations. Sales are strongly correlated with the previous week and year.

4.3.2 ADF Stationarity

ACF plots decay slowly which may indicate non-stationarity. This is addressed by applying the Augmented Dickley-Fuller (ADF) test. This reveals that *The Very Hungry Caterpillar* is non-stationary whereas *The Alchemist* is stationary. This aligns with the initial visual interpretation of Figure 1.

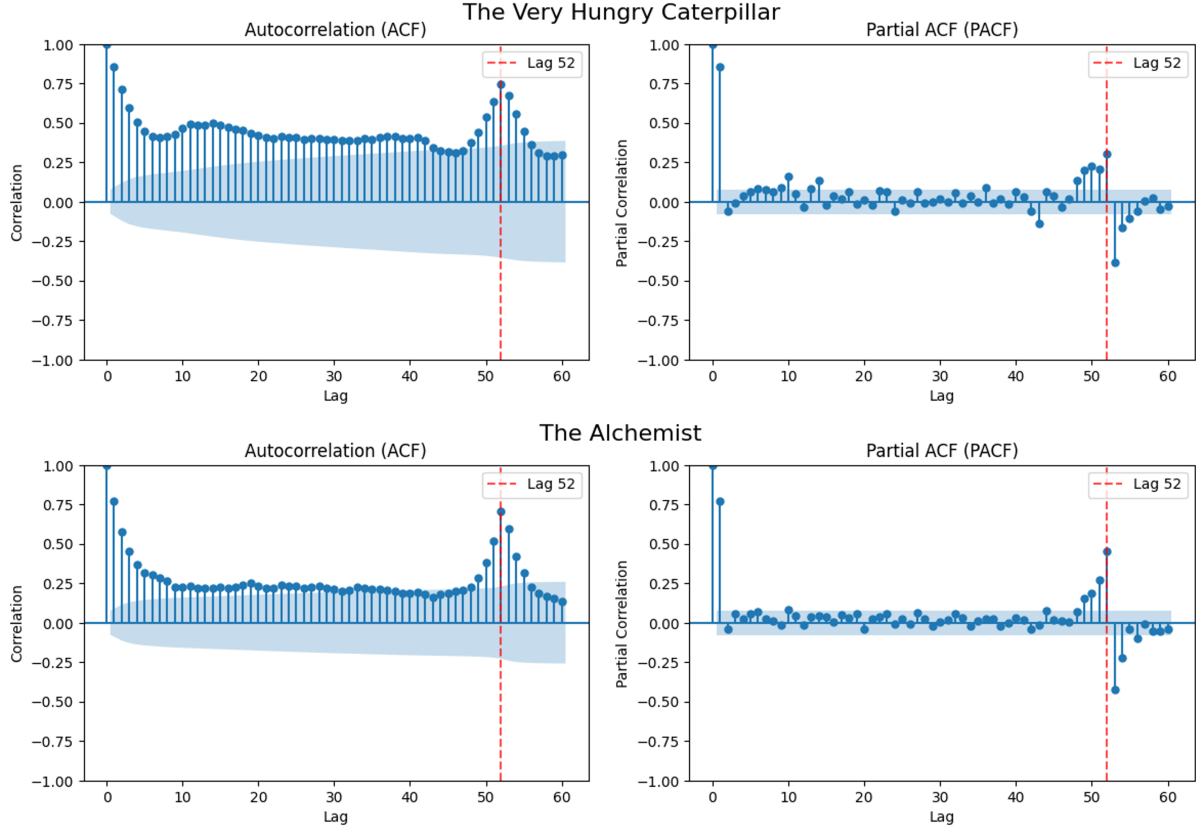


Figure 3: ACF and PACF plots for both books.

4.4 Model Performance

A standardised plotting function is used to ensure visual consistency. Auto-ARIMA applied to *The Very Hungry Caterpillar* is shown in Figure 4, whilst all results are summarised in Tables 2 and 3.

For single architecture models, classical Auto-ARIMA performed best with the lowest error scores. XGBoost and LSTM models performed worse and similarly to each other, though a more exhaustive hyperparameter tuning approach may improve this, especially for XGBoost. Auto-ARIMA’s performance and explainability make it the preferred single architecture model.

Model	The Very Hungry Caterpillar	The Alchemist
Auto-ARIMA	1466	1023
XGBoost	2149	1627
LSTM RNN	2360	967
Sequential ARIMA-LSTM	1359	1239
Parallel ARIMA-LSTM	1454	914

Table 2: MAE across tested models. Lower values denote better performance.

Model	The Very Hungry Caterpillar	The Alchemist
Auto-ARIMA	0.164	0.183
XGBoost	0.211	0.251
LSTM RNN	0.229	0.200
Sequential ARIMA-LSTM	0.150	0.233
Parallel ARIMA-LSTM	0.156	0.187

Table 3: MAPE across tested models. Lower values denote better performance.

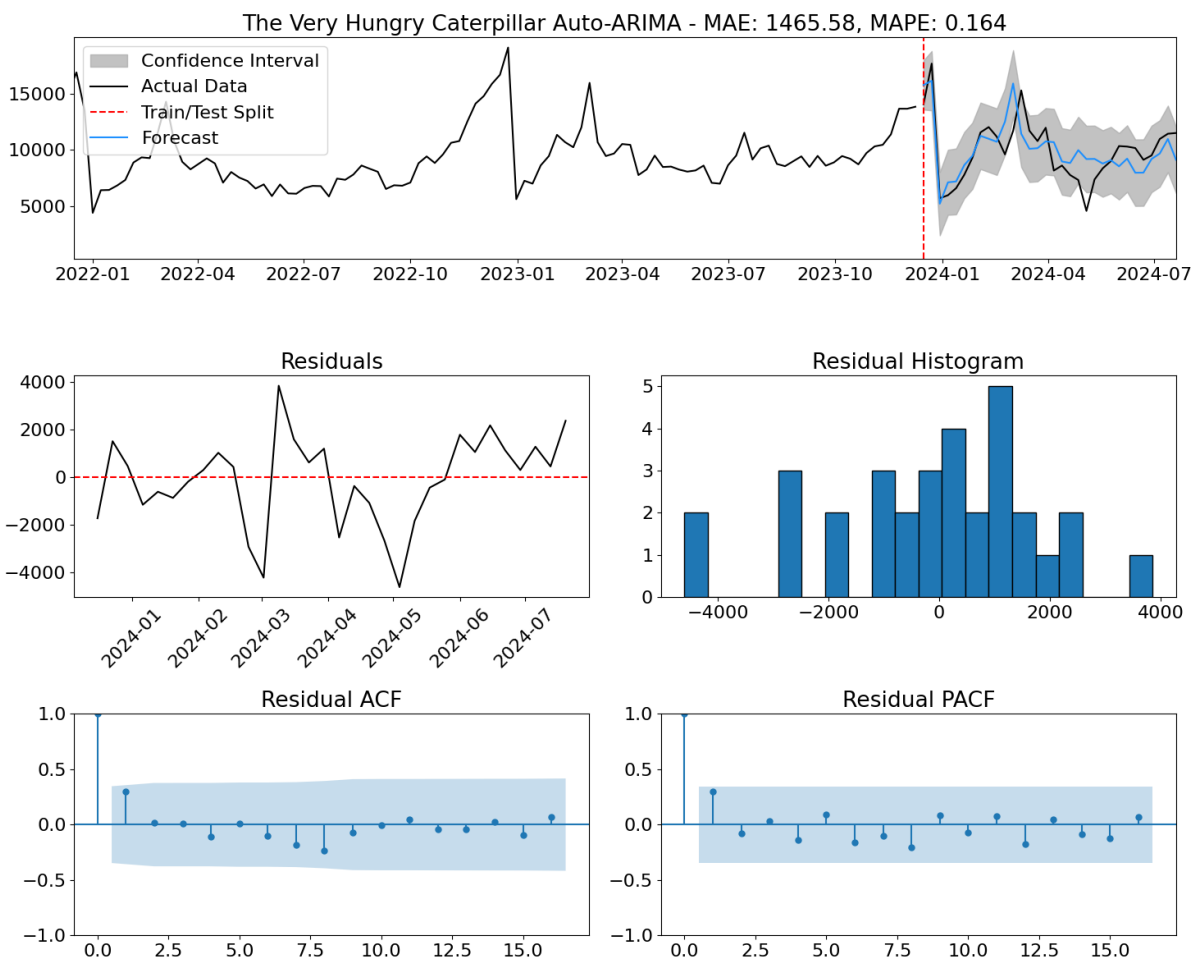


Figure 4: Forecast analysis for Auto-ARIMA applied to *The Very Hungry Caterpillar*. An eight month period is forecast with MAE and MAPE shown. The model performs well, capturing the characteristic peaks with the majority of the actual data lying within the prediction’s 95% confidence interval. Residuals behave well, centred around zero and forming a rough normal distribution. This confirms the model has captured much of the available information as residuals relate to noise. The ACF/PACF of the residuals confirm this without any statistically significant autocorrelations.

The hybrid models yielded mixed results, performing better than Auto-ARIMA for *The Very Hungry Caterpillar*, but worse than *The Alchemist*.

Sequential models gave the best model for *The Very Hungry Caterpillar*, yet the worst for *The Alchemist*. Parallel models provided strong performance across both books by optimally combining forecasts, providing the second best models for each book. *The Very Hungry Caterpillar* weighted heavily toward Auto-ARIMA, whereas *The Alchemist* showed a slight preference for LSTM. This varied weighting elegantly compensates for the mistakes individual models make at different forecast points.

Notably, the non-deterministic nature of LSTMs causes variable results. Prior runs sometimes showed Auto-ARIMA outperforming **all** models, with parallel models occasionally weighting Auto-ARIMA at 100% due to poor LSTM performance in that specific run. Similarly, previous sequential runs occasionally performed worse than a standalone Auto-ARIMA model, indicating the LSTM failed to learn anything from the residuals.

4.5 Monthly Sampling Frequency

Finally, models were also assessed on a downsampled monthly frequency, yet these performed poorly. Useful information was lost during aggregation, leading to high error scores and highlighting the need to balance information quality with computational cost.

5 Conclusion

Accurate demand forecasting is a cornerstone of proactive supply chain management and retail optimisation. This analysis demonstrates that Auto-ARIMA serves as an exceptionally strong and interpretable baseline model for book sales.

However, the findings clearly indicate that there is no universal forecasting architecture. Performance varies significantly across time series, compounded by the fluctuating performance of non-deterministic deep learning models like LSTMs. To avoid suboptimal performance with models landing in local minima, future implementations should train multiple iterations and aggregate forecasts for robustness. While Auto-ARIMA provides a solid foundation, hybrid models can offer nuanced improvements by combining the strengths of statistical, machine, and deep learning techniques based on specific data profiles.

The business impact of deploying these automated pipelines is substantial. By leveraging STL decomposition and multi-paradigm forecasting, retailers can reliably isolate long-term trends from recurring seasonal patterns. This empowers procurement teams to confidently adjust stock levels for seasonal spikes, maximising revenue capture and minimising costly inventory shortages.